

AI-NATIVE QUALITY + COMPLIANCE CLOUD · FOR LIFE-SCIENCES MANUFACTURING

An AI agent at every quality decision — from deviation to release.

Hawkeye is the system of record + system of intelligence for pharma + medical-device quality teams. One platform covers **14 EQMS modules**, the full **8-phase audit lifecycle**, supplier intelligence, regulatory readiness, and a **31-agent AI layer** that drafts, predicts, classifies, and verifies — all audit-trailed to 21 CFR Part 11.

The problem

Pharma + medical-device quality is the last unautomated knowledge function. A typical mid-market manufacturer runs:

7-12

disconnected systems for QMS · audit · supplier · doc-control · training. No single record of truth.

90+ days

average CAPA closure time. Investigations re-do work. Effectiveness checks slip.

\$1.5-3M

annual external audit + observation-remediation spend. Per FDA: average 483 = 6.2 observations.

The opportunity

The global EQMS market is **\$12B** (2025) growing to **\$24B** by 2030 (10%+ CAGR · Grand View). Audit + supplier-quality adds another **\$8B**. Adjacent regulatory-intelligence tooling is \$4B. Incumbents (MasterControl · Veeva Vault QMS · Sparta TrackWise) are pre-AI, monolithic, and on-prem-first. **FDA's January 2025 AI guidance** creates a regulator-blessed window for audit-trailed AI to enter quality. Hawkeye is built native to that moment.

The full platform — six pillars

EQMS + Audit is *one* pillar. Hawkeye is the broader cloud.

LIVE TECH SURFACE

- **92** backend route files
- **165** Mongoose models
- **14** EQMS transactional modules
- **105** click-by-click features documented
- **31** AI agents in production (Waves 1-3 + audit-agents)
- **5** public-data sources fused (openFDA · FDA WL · EMA EudraGMDP · WHO PQ · Pharma Compass)

ARCHITECTURE

- Next.js 15 · MUI · TypeScript
- Node + Express · Vercel serverless
- MongoDB Atlas · multi-tenant
- Free Gemini default · pluggable OpenAI · on-prem option

1 · EQMS Core (14 modules)

Deviation · CAPA · Doc Control · Risk · Management Review · Training · Change Control · Complaints · Batch Records · Equipment · Design Control · plus 3 supplier-side modules. Full state-machine lifecycles · role gates · 21 CFR Part 11 audit trail.

2 · Audit + Marketplace

8-phase internal audit (ISO 19011) plus a 2-sided RFQ marketplace where buyers post audit requests, auditor orgs quote, buyer awards, the awarded quote auto-converts into a full audit. Surveillance follow-ups + closure certificates with 4-role e-sig.

3 · Supplier + Regulatory Intelligence

Public-data fusion across openFDA · FDA Warning Letters · 483s · import alerts · EMA EudraGMDP · WHO PQ · Pharma Compass. Per-supplier verdict (known_tenant / public_only / ambiguous / unknown) feeds CAPA + audit decisions.

4 · AI Agent Layer (31 wired)

Wave 1 inline assists (5-Why · CAPA RCA · Deviation classifier) · Wave 2 cross-module agents (Risk Brainstormer · MRM Populator · Training Auto-Assign · Regulatory Impact Classifier) · Wave 3 governance (Predictive CAPA · Drift Monitor · Signal Detector · Active Learning Loop) · audit-agents (Supplier Intel · Audit Prep · Auditor Coach · Report Assembler with SHA-256 integrity hash).

5 · Compliance + Governance

Compliance standard registry (ISO 9001 / 13485 · 21 CFR 211 / 820 · ICH Q7 / Q9 / Q10 · EU GMP). Periodic compliance runs grade tenant readiness. Every AI decision logged with prompt-version + retrieval-set hashes for FDA AI-guidance audit trail.

6 · Multi-tenant SaaS Infrastructure

Per-tenant module flags · 7 user roles with normalised permits · electronic-signature service (PASSWORD / MFA / SSO / certificate) · notification policies · workflow milestones · DigiLocker integration for India · SubscriptionModel for pricing.

What 14 EQMS modules look like, per-module

Each module ships a state machine + CRUD + AI assists + regulator trace. Documented end-to-end in 14 separate Feature Guide PDFs (~5 MB total · 105 features click-by-click).

Quality	Audit + Supplier	Manufacturing + Design
Deviation · CAPA v2 · Doc Control · Risk Register · Management Review · Training · Change Control · Complaints	Internal Audit (8-phase) · Audit RFQ marketplace · Supplier Pre-Qualification	Batch Records · Equipment + Calibration · Design Control (med device)

Three forces converging in 2025-2026

1 · Regulator-blessed AI window opens

FDA's January 2025 final guidance on AI/ML in regulated decision-making sets the bar: prompt-version traceability, retrieval-set provenance, drift monitoring, accept/reject audit trail. Hawkeye's aiAuditTrail.js ships every one of these primitives in production today.

Source · FDA-2024-D-4488 · "Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products"

2 · Quality cost crisis at the mid-market

FDA inspection backlog post-COVID + DSCSA (Nov 2024) + EU MDR (May 2024) + ICH Q9(R1) (Jan 2024) push 5+ new audit + reporting obligations onto a quality function that is still spreadsheet-first. Mid-market sees 30-40% YoY rise in compliance cost.

Source · McKinsey "Pharma Quality 4.0" 2024 · KPMG Pharma Compliance Survey 2024

3 · Incumbents are pre-AI + on-prem

MasterControl, Sparta TrackWise, Veeva Vault QMS were architected pre-LLM. Adding AI is a re-platform. They lack live public-data fusion + agentic workflows + pluggable model providers. Switching cost is high but appetite is forming — 64% of QA leaders surveyed expect AI in QMS within 24 months.

Source · LNS Research Quality 4.0 Survey 2024 (n=412)

4 · 2-sided audit market is fragmented

Pharma audit market is **\$8B** globally; supply side is 6,000+ small auditor firms; demand side is 50,000+ manufacturing sites. No marketplace exists. Hawkeye's RFQ + 8-phase lifecycle is the only platform that runs both sides on one record.

Source · IBISWorld Pharma Audit Services 2024

Defensibility

Capability	Hawkeye	MasterControl	Veeva Vault QMS	Sparta TrackWise
Native AI agents (audit-trailed)	●	○	○	○
Public-data fusion (FDA/EMA/WHO)	●	○	●	○
2-sided audit RFQ marketplace	●	○	○	○
Pluggable LLM (free Gemini · OpenAI · on-prem)	●	○	○	○
Multi-tenant SaaS · serverless	●	●	●	○
Full EQMS (14 modules) shipped	●	●	●	●
Free-tier LLM as default (margin)	●	○	○	○
21 CFR Part 11 + ALCOA+ AI audit trail	●	●	●	●

● = full · ● = partial · ○ = absent. Based on public docs + analyst-reviewed comparisons (LNS Research · Gartner Magic Quadrant for QMS 2024).

What's shipped + verifiable

14

EQMS MODULES LIVE

105

FEATURES CLICK-BY-CLICK

31

AI AGENTS IN PRODUCTION

130 / 12 / 7

MET / PARTIAL / GAP VS
PHARMA EQMS
EXPECTATIONS

Live execution evidence (verifiable URLs + tests)

- **Live SaaS endpoints** · frontend `https://hawkeye-frontend-dev-chi.vercel.app` · backend `https://hawkeye-backend-dev.vercel.app`
- **Per-module lifecycle tests** · 11 modules walked end-to-end · **58 PASS · 0 FAIL · 0 SKIP** on live Vercel against free Gemini
- **Cross-module proof** · Deviation → AI-drafted RCA → real CAPA created in workspace + linked back · 15 PASS · 2 SKIP · 0 FAIL across 3 cross-module flows
- **AI smoke tests** · Wave 1+2+3 · 11/12 PASS · audit-agents · 9/10 PASS · all on free Gemini Flash-Lite
- **Public-data fusion live** · queries openFDA + FDA Warning Letters in production · returns ANDA records + 483s + import alerts in <3s
- **Audit-trail compliance** · every AI decision logs prompt-version + retrieval-set + provider + model + latency + accept/reject outcome (21 CFR Part 11 ALCOA+ ready)

Demo tenant · Novex Pharma Inc.

Pre-seeded with realistic FDA-483-pattern data so investors can run the platform unaided:

- 11 personas · 7 roles · single password (EqmsDemo@2026) · Universal nav
- 3 demo deviations (DEV-DEMO-001/002/003) on Novexolimus 1mg tablet · 5 FMEA risks (RPN 96-240) · 2 MRMs · 4 SOPs · 3 training records
- 1 live AI signal cluster · equipment NVX-PRESS-001 · z=3.4 (real anomaly detection)

Customer signal · pipeline

Stage	Activity	Status (as of 2026-04-25)
Design partners	Pharma + medical-device pilots	[TO FILL · # of LOIs · named pilots if disclosable]
Signed pilots	Paid 90-day pilots	[TO FILL · count · ARR per pilot]
Active ARR	Recurring revenue	[TO FILL · current ARR · MoM growth]
Auditor-side onboards	Auditor orgs on the marketplace	[TO FILL · # of auditor orgs · # of auditors]

Unit economics (model)

ACV per pharma tenant	\$24-180k (tiered by site count + modules enabled)
Gross margin	75-85% — free Gemini default + Vercel serverless keeps inference cost negligible at scale
Marketplace take-rate	8-12% on RFQ award value (auditor-side)
Net retention target	120%+ via module expansion + audit-marketplace usage

12-month product roadmap

Q2 · 2026 (IN FLIGHT)

Polish + paid pilots

- E-sig enforcement on closures (CAPA · MRM · batch · doc)
- OVERDUE schedulers across modules (notificationOutbox)
- FAR / MDR clock + auto-reminder
- Full UI for CAPA-v2 5-stage approval gates

Q3 · 2026

Tenant ROI controls + new verticals

- AI ROI / token-budget dashboard for tenant admins — choose free Gemini · paid OpenAI · on-prem; per-tenant budgets + per-user caps
- Medical-device vertical pack (ISO 13485 · MDR / IVDR)
- AI similarity / recurrence detector across CAPA history
- DHF auto-assembler on Design Control transfer

Q4 · 2026 — Q1 · 2027

Adjacent markets + integrations

- Food + beverage vertical pack (FSMA / HACCP / ISO 22000)
- FDA MedWatch eMDR submission API
- Sigstore / blockchain anchoring for audit-report integrity hashes
- Native ERP connectors (SAP S/4 · Oracle Fusion · NetSuite)
- Auditor org marketplace billing + payouts

The ask

Funding request to harden the product, sign 12-25 design partners into paid pilots, and unlock the auditor-side marketplace.

[TO FILL · \$X.X M]

Round size · Seed / Pre-A

[TO FILL · 18-24 mo runway]

Use of funds · 60% engineering · 25% GTM · 15% compliance

[TO FILL · lead investor]

Co-lead + 2 strategic angels reserved

Team

[TO FILL · Founder name]

Founder & CEO

Pharma quality + SaaS background · prior exits · led platform architecture from zero to live SaaS in 2026.

[TO FILL · Co-founder / advisors]

CTO / Advisors

Pharma QA leaders + ex-FDA + AI infra; advisor list available under NDA.

Why this team will win

- Velocity.** 14 modules + 31 AI agents + cross-module workflows + live SaaS + 105 click-by-click features documented + 58-step lifecycle test suite — all green on live infra. Most pre-seed teams ship 1 module.
- Regulator literacy.** Every state machine cites the source CFR / ISO / ICH clause. AI audit trail meets the FDA Jan 2025 guidance bar today.
- Cost discipline.** Free-tier Gemini as default keeps gross margin in the 80s while the customer pays for paid model upgrades. Most AI-quality competitors are buying token credits at retail.
- Two-sided distribution.** Auditor-side onboarding builds a moat that buyer-only EQMS can't copy. Auditors bring their buyer relationships.